

# ADARSH PATEL

AI ENGINEER | FULL STACK DEVELOPER

+91-6267597624 • eng.adarshpatel@gmail.com • [linkedin.com/in/adarshpatelaiengineer](https://www.linkedin.com/in/adarshpatelaiengineer) • Delhi, India

## PROFESSIONAL SUMMARY

AI Engineer with 3+ years of experience designing enterprise-scale Generative AI and Agentic AI platforms using LangChain, LangGraph, FastAPI, and AWS cloud-native architectures. Specialized in multi-agent orchestration, RAG pipelines, NLQ-to-SQL engines, multimodal AI systems, and no-code AI agent platforms enabling organizations to automate analytics, knowledge discovery, and business workflows. Architected a production-grade AI Agent Builder platform allowing teams to visually create and configure AI agents for analytics, document intelligence, video intelligence, and enterprise automation using OpenAI and Gemini LLM models with MCP-style tool integrations. Strong experience in AI infrastructure, distributed microservices, vector databases, and scalable cloud deployments on AWS ECS/Fargate.

## TECHNICAL SKILLS

|                |                                                                                                                                                                          |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI / ML        | LangChain • LangGraph • RAG Pipelines • Multi-Agent Orchestration • OpenAI • Gemini • Hugging Face • TensorFlow • PyTorch • Pinecone • NLQ-to-SQL • MCP Tool Integration |
| Languages      | Python • JavaScript • Java • SQL                                                                                                                                         |
| Backend        | FastAPI • Flask • Spring Boot • Node.js (Express) • REST APIs • WebSockets • JWT • SSO • RBAC • Microservices                                                            |
| Frontend       | React.js • Tailwind CSS • Bootstrap • AG-Grid • AG-Charts • Module Federation • Webpack                                                                                  |
| Cloud / DevOps | AWS ECS/Fargate • EC2 • S3 • RDS • Lambda • CloudFront • ALB • Route 53 • Docker • GitHub Actions • CodePipeline • CodeBuild • Nginx                                     |
| Databases      | PostgreSQL • MySQL • MongoDB • Redis • Snowflake • AWS Redshift • SQLAlchemy • Hibernate                                                                                 |

## PROFESSIONAL EXPERIENCE

### Full Stack Developer (AI Engineer)

**BITTWOBYTE Private Limited** • Rewa, India

May 2023 – Present

#### Artificial Intelligence Integration

- Architected an enterprise-scale Agentic AI platform with LangGraph-based multi-agent orchestration, enabling autonomous agents to collaborate across analytics, document intelligence, video intelligence, and enterprise integrations.
- Designed and built a No-Code AI Agent Builder and voice/chat-powered AI Copilot, enabling users to visually create domain-specific workflows and control the platform via speech or text.
- Built an NLQ-to-SQL engine converting natural language queries into optimized SQL with real-time visualizations, improving data insights by 70%.
- Developed RAG-based knowledge intelligence systems supporting document ingestion, chunking, embeddings, semantic search, and contextual AI responses.
- Implemented MCP-style tool integrations and real-time voice AI using Gemini Live API and WebSocket streaming, enabling agents to interact with enterprise APIs, CRM systems, and databases.
- Integrated AI agents with third-party applications and messaging channels (WhatsApp, Slack), enabling seamless end-user interaction and remote workflow execution.

- Designed vector memory architecture using Pinecone and Redis, improving AI response accuracy by 70% and reducing token consumption by 20%.
- Developed KPI Wallet for pinning NLQ insights as live KPI cards and built an AI-powered Resume-JD matching engine for ATS-style skill gap analysis.
- Built AI Tutor — an adaptive education platform with AI Student Personality Engine, delivering personalized LLM-driven tutoring, intelligent assessments, and adaptive learning workflows.

### Backend Development

- Built scalable RESTful APIs using FastAPI, Flask, Spring Boot, and Express powering high-performance microservices architectures.
- Implemented JWT authentication, Microsoft SSO with Spring Security, and multi-tenant RBAC architecture ensuring secure isolated data access.
- Reduced API latency by 40% using Redis caching, async processing, and connection pooling optimizations that cut service startup time by 30%.
- Designed and implemented WebSocket-based real-time communication for streaming AI responses and live notifications.
- Implemented API rate limiting, throttling, and token-based usage tracking to manage enterprise-level API consumption.

### Cloud & DevOps

- Designed and provisioned complete AWS cloud infrastructure for microservices-based architecture; containerized services with Docker and deployed on AWS ECS/Fargate with ALB Auto Scaling and CloudFront CDN.
- Built CI/CD pipelines using GitHub Actions, AWS CodePipeline, and AWS CodeBuild enabling automated testing, build, and deployment workflows.
- Delivered end-to-end production support including incident response, debugging, and hotfix deployment, improving recovery speed by 50%.

### Frontend Development

- Developed responsive UIs using React.js, Tailwind CSS, and Bootstrap; integrated AG-Grid and AG-Charts delivering KPI dashboards that reduced reporting dependencies by 70%.
- Configured micro-frontends with Module Federation enabling independently deployable UI modules.

### Database Management

- Designed scalable multi-tenant database architecture using MySQL/PostgreSQL/Redshift ensuring secure tenant-level data isolation.
- Enhanced DB performance by 60% via indexing, query optimization, and Hibernate tuning; developed stored procedures for complex business logic.
- Improved large-scale processing by optimizing Snowflake workflows, reducing runtime by 35%.

## Software Developer Intern

TECHLOZY · Rewa, India

Oct 2022 – Mar 2023

- Developed full-stack applications using React.js, Node.js, and Express, improving system scalability and responsiveness.
- Built REST APIs for authentication and data operations, improving backend efficiency by 25%.
- Integrated MongoDB and PostgreSQL with optimized schemas, improving query performance.
- Deployed cloud applications using AWS EC2, S3, and Docker with CI/CD running on GitHub Actions.

## KEY PROJECTS

### AI Multi-Agent Automation Platform

#### Team Lead / Full Stack Developer

React | Python FastAPI | LangChain | LangGraph | LLM Models | ElevenLabs TTS | Pinecone | Redis | WhatsApp API | Slack API | Docker | AWS ECS | CloudFront | Nginx

- Built a production-grade multi-agent AI platform with a No-Code Agent Builder for NLQ-to-SQL analytics, RAG knowledge systems, video intelligence, and multimodal AI workflows.
- Developed an integrated voice and chat-powered AI Copilot enabling users to build agents, trigger workflows, and fully control platform actions via natural language.

- Integrated AI agents with WhatsApp and Slack messaging channels, enabling users to trigger workflows and query agents via chat.
- Implemented RBAC, Microsoft SSO, and AWS auto-scaling ensuring reliable, secure multi-tenant deployment.
- Delivered measurable business impact: 75% faster reporting workflows and 80% reduction in document search time.

## AI Tutor — Adaptive AI-Powered Education Platform

### Full Stack Developer / AI Engineer

*React | Python FastAPI | LangChain | LangGraph | LLM Models | ElevenLabs TTS | Redis | Docker | AWS ECS | CloudFront*

- Built a full-stack adaptive AI tutoring platform with Admin, Teacher, and Student dashboards, delivering personalized LLM-driven learning experiences.
- Developed an AI Student Personality Engine analyzing learning styles, cognitive proficiency, and knowledge gaps to generate personalized tutoring strategies.
- Built a custom RAG pipeline leveraging Gemini embeddings and semantic retrieval for context-aware tutoring from uploaded PDF/Text materials.
- Implemented AI-powered Quiz and Lesson Plan Generator with automated remedial learning system for gap analysis and targeted follow-up quizzes.

## Laboratory Information System (LIS)

### Backend Developer

*Java | Spring Boot | Thymeleaf | MySQL | REST APIs | Hibernate | AWS*

- Built REST APIs for lab operations, patient handling, and automated reporting; developed Thymeleaf UI improving technician efficiency.
- Implemented Spring Security meeting healthcare-grade compliance; deployed on AWS with automated CI/CD pipelines, ensuring 99.9% system uptime.

## School Management System ERP (B2B)

### Full Stack Developer

*React Vite | Node.js Express | PostgreSQL | Module Federation | Microservices | JWT | AWS | Nginx*

- Designed multi-tenant microservices architecture for a B2B ERP system with secure JWT authentication, role-based dashboards, and containerized deployment using Docker, ECS, CloudFront, and Nginx.

## EDUCATION

---

### Bachelor of Technology in Computer Science Engineering

RGPV University, Bhopal, Madhya Pradesh, India

2023

## CERTIFICATIONS

---

- ▶ AWS Certified Solutions Architect – Associate
- ▶ AWS Certified Cloud Practitioner

## LANGUAGES

---

**English:** Fluent • **Hindi:** Native